

Basic Counting, Restrictions & Inclusion-Exclusion

Integrated Grade-9 learner pack

Basic Counting, Restrictions & Inclusion-Exclusion

Integrated Assimilation Book

Count objects, not formulas. First decide what makes two outcomes the same.

1. RECONNECT - what is the counted object?

Suppose you own 3 shirts and 4 caps. An outfit is an ordered pair (shirt, cap). There are $3 \times 4 = 12$ outfits because every shirt choice can be followed by every cap choice.

Now suppose you may travel by one of 5 buses or one of 3 trains. These are disjoint types of journey, so there are $5 + 3 = 8$ choices.

The symbols $+$ and $*$ are not automatic. They encode different structures:

- add counts from disjoint alternatives;
- multiply counts sequential stages when every first-stage choice can be combined with the counted second-stage choices.

The exact-one rule

Before adding case counts, ask:

Does every valid object enter exactly one case?

If some object enters two cases, naive addition double-counts. If some object enters no case, the split is incomplete.

2. DISCOVER - ordered or unordered?

Choosing a captain and vice-captain from 8 people is ordered: (A,B) and (B,A) are different roles. Count $8 \times 7 = 56$.

Choosing two representatives from 8 is unordered: {A,B} and {B,A} are the same pair. The ordered count 8×7 counts each pair twice, so divide by $2!$:

$$C(8,2) = 8 \times 7 / 2 = 28.$$

This is the meaning behind permutation and combination notation:

- order matters -> ordered arrangements;
- order does not matter -> remove the overcount caused by rearranging the chosen objects.

3. MAKE SENSE - derive, do not memorize

For r ordered selections from n distinct objects without replacement:

$$n \times (n-1) \times \dots \times (n-r+1).$$

For an unordered r -subset, each chosen set appears in $r!$ orders, so

$$C(n,r) = n \times (n-1) \times \dots \times (n-r+1) / r!.$$

The formula is the last line of the reasoning, not the first question to ask.

4. Repeated objects change identity

How many distinct arrangements of AABC are there?

If the two A's were temporarily named A1,A2, there would be 4! arrangements. Swapping A1,A2 changes no visible word, so every visible arrangement was counted 2! times.

Therefore the count is $4!/2!=12$.

For multiplicities $m_1, m_2, \dots$ summing to n , the same identity argument gives

$n!/(m_1!m_2!\dots)$.

Use this only when objects within each repeated class are genuinely indistinguishable for the problem.

5. Restrictions should enter before the factorial

To form a 4-digit odd number using 1,2,3,4 exactly once, the units digit must be 1 or 3. Choose the restricted position first:

$$2 * 3! = 12.$$

This is often cheaper than counting all 4! permutations and then testing parity.

Position-first cue

When one condition strongly controls a position (last digit, first digit, fixed role, forbidden envelope), handle that position or restriction before expanding the rest of the count.

6. Complement - count the easier opposite event

To count length-7 binary strings with at least one 1, direct cases by first 1 are possible but unnecessary.

All binary strings: 2^7 .

The only forbidden string with no 1: one string.

Required count: $2^7-1=127$.

Complement is especially useful for phrases such as:

- at least one;
- not all;
- at least one violation;
- a difficult relative-order event whose opposite has rigid structure.

7. Inclusion-exclusion - overlap must be repaired explicitly

Among the integers 1,...,100, how many are divisible by 2 or 5?

Multiples of 2: 50.

Multiples of 5: 20.

Multiples of both, i.e. 10: 10.

If we add $50+20$, the multiples of 10 were counted twice. Subtract the overlap once:

$$50+20-10=60.$$

For three properties, the same principle continues:

single counts - pair overlaps + triple overlap.

Do not use inclusion-exclusion when the original cases are already disjoint.

8. Digit strings: counting structure versus arithmetic structure

A digit problem can belong to two different worlds.

Counting question: How many 4-digit odd numbers use 1,2,3,4 exactly once? The arithmetic condition only restricts the last position; the main work is counting admissible arrangements.

Arithmetic digit question: Which numbers are divisible by 11, or how does a decimal block reduce modulo a divisor? That arithmetic mechanism belongs to the later digit-structure topic. Here we may retrieve such a restriction, but we do not reteach its number-theory derivation.

Boundary rule:

If the property is already known and the task is to count strings satisfying it -> count here.

If the task is to derive or exploit the arithmetic digit property itself -> route to number theory.

9. Historical anchors as decision models

Restricted digit relation

The 2025 three-digit anchor is not a permutation problem. The relation $c=a+b$ converts the number into bounded integer-pair choices. Counting admissible (a,b) pairs gives 45.

Grouped restricted assignment

The 2025 coupon anchor treats (1,2), (3,4), (5,6) as three grouped objects assigned injectively to six envelopes, each with two forbidden envelopes. The restriction belongs in the assignment model before any factorial shortcut. The verified count is 40.

Multiset + complement

For permutations of 223334444, repeated digits require multiset counting. The relative-order condition becomes simpler when phrased through the last symbol among the 3/4 subsequence; the verified count is 540, giving remainder 40 modulo 100.

Position restriction

The 2024 odd-number anchor chooses the units digit first, then permutes the remaining symbols: 12.

Rotational equivalence

The 2023 dice anchor first fixes the opposite 1,2 axis. The remaining four labels have six cyclic orders around that axis; the three opposite face-pairs each choose one of two colours. The repository's independent verification gives 48. No formal group-action machinery is needed for this Grade-9 interface.

Unordered selection symmetry

The 2023 order-statistic anchor treats increasing 5-tuples as 5-subsets. Reflection/order-statistic symmetry avoids enumerating every subset and leads to 66.

Finite-set constraints

The 2023 finite-set anchor first solves constraints on set size and maximum, then uses binomial counts for lower elements. It is a model-first count, not raw subset enumeration; the verified requested value is 43.

10. TRY - support fades toward independence

Full support

How many 5-bit strings contain exactly two 1s?

Identify the two positions of the 1s: $C(5,2)$.

Medium support

How many 4-person committees from 10 contain at least one of two specified people?

Cue: the complement excludes both specified people.

Light support

How many arrangements of AABBC have the two A's nonadjacent?

Cue: total minus an AA block.

Independent

How many 6-letter strings over $\{A,B,C,D\}$ use every symbol at least once?

Choose and justify direct/complement/inclusion-exclusion without a method label.

11. DIAGNOSE - common counting failures

- Object drift: switching halfway from ordered sequences to unordered sets.
- Case overlap: adding counts when one object satisfies two cases.
- Case gap: a split does not cover every valid object.
- Factorial reflex: writing $n!$ before restrictions are modeled.
- Repeated-object overcount: treating identical copies as labelled.
- Complement mismatch: subtracting from the wrong universe.
- Arithmetic/counting collision: deriving a divisibility rule inside a problem whose actual task is only to count already-characterized strings.

12. ADOPT - the counting checklist

1. What exactly is one object/outcome?
2. When are two outcomes considered the same?
3. Is order structural?
4. What restrictions must every object satisfy?
5. If splitting into cases, are the cases disjoint and exhaustive?
6. If making stages, does each stage count the choices available after previous choices?
7. Would the complement be simpler?
8. If properties overlap, where is inclusion-exclusion needed?
9. Are repeated symbols genuinely indistinguishable?
10. Does a digit condition belong to counting or to arithmetic structure?
11. Check small cases or an alternate count when practical.

13. TRANSFER

This language is designed to be retrieved downstream.

For recurrence/state problems, the consumer may say: define the counted state, split every valid object into exactly one first-step branch, multiply within stages, and add only disjoint branches. It should not re-teach combinations, repeated-object formulas or inclusion-exclusion.

For graph/coloring problems, the same counted-object, restriction and complement vocabulary applies, while graph-specific canon remains with its owner.

Basic Counting - First-Step Reference

One-question router

What is one counted object, and when are two objects the same?

Then ask:

ORDER? -> RESTRICTIONS? -> DISJOINT CASES OR STAGES? -> DIRECT / COMPLEMENT / IE -> CHECK

Recognition atlas

Visible clue | First move

one choice followed by another | define stages; multiply stage counts

one of several nonoverlapping types | prove disjointness; add case counts

roles/sequence/positions | treat as ordered unless the problem identifies orders

committee/subset/group | treat as unordered unless roles are later assigned

repeated symbols | divide labelled count by permutations of identical copies

last/first digit restriction | handle the restricted position first

at least one / not all | test whether complement is shorter

two properties joined by "or" | check overlap before adding

multiple overlapping restrictions | consider inclusion-exclusion

digit condition | decide whether arithmetic property is given or must be derived

Exact-one case test

Before adding branches:

1. Every valid object belongs to at least one branch.
2. No valid object belongs to two branches.

If either fails, repair the split before counting.

Identity test

Before dividing by a factorial, state exactly which labelled arrangements become the same visible object.

Basic Counting - Recognition and First-Line Lab

Write only the first useful model/counting line unless asked for a value.

1. Choose a captain and vice-captain from 9 students: ordered or unordered?
2. Choose a 3-person committee from 9: ordered or unordered?
3. A meal chooses one starter from 4 and one main from 6: add or multiply?
4. A route is either one of 5 bus routes or one of 2 train routes: what condition makes addition valid?
5. Arrange AABC: what overcount must be removed?
6. Form a 4-digit even number using 1,2,3,4 once: which position should be chosen first?
7. Count 8-bit strings with at least one 1: name the easier complementary event.
8. Count integers divisible by 3 or 5: what overlap must be corrected?
9. Split a count by "starts with A" and "contains A": why is naive addition unsafe?
10. A recurrence splits tilings by first tile: what exact-one question must be checked before adding branches?
11. Assign three distinct tasks to five people, at most one task per person: identify the sequential stages.
12. Select 4 people from 10 with at least one of two specified people: write the complement universe.
13. A word contains repeated L's and O's: what must be decided before using a multiset formula?
14. A number must be divisible by 4 and the divisibility characterization is already supplied: is the remaining task arithmetic derivation or counting?
15. Two restrictions overlap: state when inclusion-exclusion becomes necessary.
16. A set problem specifies the maximum element: what must every counted subset contain?

Basic Counting - Practice and Transfer Bank

Foundation

1. An outfit uses one of 3 shirts and one of 4 caps. How many outfits?
2. A trip uses exactly one of 5 bus routes or 3 train routes. How many route choices?
3. How many 2-person committees can be chosen from 6 people?
4. How many linear arrangements of 4 distinct books are possible?
5. How many distinct arrangements of the letters AABC are possible?

Direct

6. How many 4-digit even numbers can be formed using 1,2,3,4 exactly once?
7. How many 3-person committees can be chosen from 8 people?
8. How many distinct arrangements of BANANA are possible?
9. How many length-3 strings can be formed from 1,2,3,4,5 without repetition?
10. How many binary strings of length 5 contain exactly two 1s?

Standard

11. How many 4-digit numbers can be formed from 0,1,2,3,4,5 without repetition?
12. In how many permutations of 1,2,3,4,5,6 does 1 occur before 2?
13. How many subsets of $\{1,2,3,4,5,6,7,8\}$ contain exactly one of 1,2?
14. Three distinct tasks are assigned to three different people chosen from five. How many assignments?
15. How many length-3 strings over $\{0,1,2,3,4\}$ contain at least one 0?

Disguised

16. How many distinct arrangements of BALLOON are possible?
17. How many 5-digit even numbers can be formed from 0,1,2,3,4,5,6 without repetition?
18. How many 4-person committees from 10 contain at least one of two specified people?
19. How many permutations of 1,2,3,4,5,6,7 have 1 and 2 nonadjacent?
20. Five people stand in a line. In how many orders is A left of B and C left of D?
21. Four letters are placed into four addressed envelopes, one per envelope. How many placements put no letter in its matching envelope?
22. How many integers from 1 through 100 are divisible by 2 or 5?

Preliminary-style

23. How many length-6 strings over $\{A,B,C,D\}$ use every symbol at least once?
24. How many distinct arrangements of the multiset 112233 end in 3?
25. How many distinct length-7 strings with three 1s, two 2s and two 3s do not begin with 1?
26. A committee of 5 is chosen from 12 people, including 4 designated seniors. How many committees contain at least 2 seniors?
27. In a class of 30, 18 study mathematics, 16 study physics and 9 study both. How many study at least one of the two subjects, and how many study neither?
28. How many length-5 digit strings (leading zero allowed) use distinct digits from 0,1,2,3,4,5 and contain exactly two even digits?
29. How many distinct arrangements of AABBC have the two A's nonadjacent?
30. How many length-6 strings over $\{0,1,2\}$ use all three symbols at least once?

Basic Counting - Independent Mixed Mastery Check

No default hints or method labels.

1. Choose a captain and vice-captain from 8 students. How many choices?
2. Choose two representatives from 8 students. How many choices?
3. How many 4-digit odd numbers use 1,2,3,4 exactly once?
4. How many distinct arrangements of LEVEL are possible?
5. How many length-7 binary strings contain at least one 1?
6. How many integers from 1 through 120 are divisible by 3 or 4?
7. How many 4-person committees from 9 do not contain both A and B?
8. How many 5-digit numbers can be formed from 0,1,2,3,4,5 without repetition?
9. Four letters are placed into four addressed envelopes, one per envelope. How many placements have no correct address?
10. How many distinct arrangements of 112233 end in 2?
11. A committee of 4 is chosen from 10 people. How many contain at least one of three specified people?
12. How many permutations of A,B,C,D,E,F satisfy A before B, C before D and E before F?
13. How many length-5 strings over {A,B,C} use all three symbols?
14. How many distinct arrangements of BANANA are possible?
15. How many length-5 digit strings (leading zero allowed) use distinct digits from 0,1,2,3,4,5 and contain exactly two even digits?
16. How many 3-person committees from 8 contain at least one of two specified people?